

Element C:
25" deep
9' 4" wide
C1:

- Must be able to hold 5-6 pieces of pottery that have a density of 2-5 g/cm³
- Shelf needs to be lighter than 23 lbs
- Each shelf must not be longer than 45"
- Each shelf must be stable and flex
- Total row of shelves must not be more than 9' 4"
- The shelves must not have a width of more than 25"
- Should not look "tacky"
- should look "fun and artsy" (Needs to not be boring looking)

c2-c3-c4-c5

Priority	Criteria	Explanation	Source	What is Measured
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1.Ms. Weaver approved because she needs to put around 5 or 6 pieces of pottery on the shelf so this is an important criteria.

1. High	Must be able to hold 5-6 pieces of pottery that have a density of 2-5 g/cm ³	The shelf is in the art showcase and needs to support heavy pottery. Pottery is a dense material and the shelf needs to be very sturdy and strong so that it does not fall or break leading to the pottery shattering.	Mrs. Weaver	The amount of weight the shelf needs to withstand. If the shelf can hold the pottery.
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2.Ms. Weaver approved because the glass is currently 23lbs causing the aluminum supports to bend. So this is an important criteria to meet to

2.High	Shelf needs to be lighter than 23 lbs	The aluminum shelf supports are bending and due to the weight of the heavy glass shelf. The glass shelf weighs 23 lbs so the new shelves that we create must weigh less than 23 lbs in order to take strain off of the aluminum supports.	Taking measurements from the showcase.	The weight of the shelf is measured. If the aluminum supports can hold and withstand the weight of the shelves we make.
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3.High	Total row of	The showcase is	Taking	The length
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protect the pottery. 3. Ms. Weaver approves because the size of the showcase is only 9' 4" long so it is important that the shelves fit in the showcase. 4. Ms. Weaver approves because from the back of the showcase to the front it is 26' deep so we need to make them less than 26' so the shelving units can fit. This is an important criteria so we can fit all the shelves. 5. Ms Weaver approved because the current shelves are 90' long making them bend		shelves must not be more than 9' 4"	also only 9' 4" long so having a row of shelves that is longer than that will cause them to not fit in the showcase. Our new product must be shorter than 9' 4" in order to be fully effective because otherwise it will not fit in the showcase and not be able to work.	measurements from the showcase.	of the shelves are measured. If the shelves can fit inside the showcase.
	4.High	The shelves must not have a width of more than 25'	From the back of the showcase to the front it is 26' deep. Therefore if we made the shelves any more than 25' wide it would not fit properly into the showcase. If it does not fit in the showcase the shelves will fail to serve a purpose. So our new product must be thinner than 25' in order to work.	Taking measurements from the showcase.	The width of the shelf. If the shelf is able to fit in the showcase. The showcase is only 25' deep so the shelf must not be that wide.
	5.High	Each shelf must not be longer than 45"	The current shelves are 90" but since they are so long the weight isn't distributed properly. With the weight not being distributed properly it makes the glass prone to breaking. So if we break the shelves up into smaller segments we can distribute the pottery better and make the glass less prone to breaking. Making the shelves smaller will make	Measurements from showcase	The length of each shelf. They must not be longer than 45". Also the weight that the shelf can withstand. Making the shelf shorter allows it to hold more weight without bending.

because the weight is not distributed properly. This criteria is important so the shelves don't break. 6. Ms. Weaver approved because she is afraid of the shelves shattering due to their lack of flex. She thinks it is a good idea to have a material that will bend rather than shatter as soon as too much weight is applied to it. 7. Ms. Weaver approved of this criteria because she is all about creativity and fun artsy look. However, she is an opinionate			them much safer due to the weight being properly distributed meaning the shelves won't break as easily.		So if the shelf can hold high weight.
	6.Low	Each shelf must be stable and can flex	The shelves in the showcase are made of glass which are non-flex and will shatter when too much weight is put upon it. This is very dangerous and will break the pottery. A flexible material such as wood will not shatter and therefore not be dangerous or break the pottery as easily when too much weight is put upon it.	Mrs. Weaver and glass research	The physical properties of the material we make the shelf out of. If the material we use can withstand force and bend and flex rather than shatter
	7.Low	The shelf should look fun and artsy but not "tacky"	It is going into the art showcase and needs to look decent and not take away from the art itself. However, it is part of the art department so it should reflect the creative and artsy nature of the department it is representing.	Ms. Weaver	Appearance

d person and critical so she doesn't want it to look tacky.		
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